Practice workbook · Finding the median

Finding or describing the middle value of ordered data.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: PSY-RM-16

1. Question 2 · part d · 2 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part d only.

2 Jenna investigated whether stress affected performance on a memory test.

She recruited 10 participants from her university. Participants were asked to remember a list of 20 words. Fifteen minutes later, they were asked to write down as many words as they could remember. The same participants were tested twice:

- Test 1: one week before their examinations began.
- Test 2: one week after their examinations had finished.

(a) Describe how Jenna could use an opportunity sampling method to gather the participants for her investigation.

(2)

(b) State which experimental research design Jenna used in her investigation.

(1)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

1. Question 2 · part d · 2 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part d only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Jenna recorded the total number of correct words remembered by participants both before and after their examinations. Her results are shown in **Table 2**.

Participant	Total number of correct words remembered one week before examinations began (out of 20)	Total number of correct words remembered one week after examinations finished (out of 20)
A	5	8
B	6	7
C	7	9
D	4	8
E	5	7
F	3	7
G	9	9
H	10	11
I	6	6
J	5	8

Table 2

(c) Calculate the mean score for the total number of correct words remembered one week **before** the participants' examinations began.

(1)

Mean .....

(d) Calculate the median score for the total number of correct words remembered one week **after** the participants' examinations finished.

(2)

Median .....

(Total for Question 2 = 6 marks)

1. Question 2 · part d · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 5 · part whole · 2 marks

2019 JUN · Paper 02

Further practice · Attempt part whole only.

DO NOT WRITE IN THIS AREA

5 Describe how you would find a median score.

(Total for Question 5 = 2 marks)

Working and answer

OLevels.com SCHOOL

3. Question 2 · part b · 2 marks

2020 NOV · Paper 02

Further practice · Attempt part b only.

2 Michelle conducted a study to find out if 15 year olds had better long-term memory recall than 55 year olds. She gave two groups of participants one hour to memorise a set of 40 pictures. After two weeks, she asked the participants to recall as many of the pictures as they could.

Her results are shown in **Table 1**.

15 year olds	Number of pictures recalled correctly	55 year olds	Number of pictures recalled correctly
Participant A	35	Participant A	16
Participant B	30	Participant B	28
Participant C	29	Participant C	27
Participant D	18	Participant D	34
Participant E	38	Participant E	35
Participant F	31	Participant F	26
Participant G	24	Participant G	21

Table 1

(a) Calculate the mean score for the number of pictures recalled by 55 year olds.

You must give your answer to two decimal places.

(2)

Mean score for 55 year olds .....

(b) Calculate the median score for the number of pictures recalled by 15 year olds.

(2)

Median score for 15 year olds .....

(c) Calculate the range for the number of pictures recalled by all participants.

(1)

Range for all participants .....

(Total for Question 2 = 5 marks)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

3. Question 2 · part b · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 4 · part c · 2 marks

2022 JUN · Paper 02

Further practice · Attempt part c only.

4 Jamie investigated whether personality influences obedience to an authority figure. He used a questionnaire with open-ended and closed-ended questions about personality traits and obedient behaviour.

(a) Describe **one** reason why Jamie may have used open-ended questions in his investigation.

(2)

(b) State **one** ethical issue that Jamie should have considered in his investigation.

(1)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

4. Question 4 · part c · 2 marks

2022 JUN · Paper 02

Further practice · Attempt part c only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Jamie found that eight of his participants scored highly for authoritarian personality.

The number of obedient acts they had reported for the previous week are shown in **Table 4**.

Participant	Number of reported obedient acts (in a week)
A	15
B	14
C	18
D	16
E	18
F	14
G	13
H	19

Table 4

OLevels.com SCHOOL

(c) Calculate the median score for the number of obedient acts reported by participants in **Table 4**.

(2)

Median .....

(d) Calculate the range for the number of obedient acts reported by the participants in **Table 4**.

(1)

Range .....

(Total for Question 4 = 6 marks)

4. Question 4 · part c · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 3 · part b · 2 marks

2023 JUN · Paper 02

Further practice · Attempt part b only.

3 Quentin is investigating the capacity of short-term memory. He showed 20 objects to 10 participants for 30 seconds. Quentin then asked them to recall as many as they could.

His results are shown in **Table 2**.

Participant	Number of objects recalled
A	10
B	9
C	8
D	8
E	6
F	7
G	8
H	9
I	10
J	11

Table 2

(a) Calculate how many participants recalled nine objects as a fraction of all participants.

You must give your answer in the lowest form.

(1)

Fraction .....

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

5. Question 3 · part b · 2 marks

2023 JUN · Paper 02

Further practice · Attempt part b only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(b) Calculate the median for the number of objects recalled.

(2)

Median .....

(c) Calculate the mode for the number of objects recalled.

(1)

Mode .....

(d) State how Quentin would calculate the mean score for the number of objects recalled.

(1)

.....

.....

.....

(Total for Question 3 = 5 marks)

OLevels.COM — SCHOOL —

5. Question 3 · part b · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 5 · part e · 2 marks

2024 JUN · Paper 02

Further practice · Attempt part e only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

5 Rocio conducted a case study on her patient who had an addiction to alcohol. She worked with the patient for six months to help treat their addiction.

(a) Define what is meant by a 'case study'.

(1)

(b) Rocio recorded the conversations with the patient as part of her case study. She also monitored the alcohol consumption of the patient in units of alcohol (one unit of alcohol = 10 ml).

Her results are shown in **Table 2**.

Month	Total alcohol consumption in units
March	298
April	221
May	181
June	140
July	99
August	59

Table 2

Describe, using an example from Rocio's case study, what is meant by 'quantitative data'.

(2)

OLevels.COM SCHOOL

6. Question 5 · part e · 2 marks

2024 JUN · Paper 02

Further practice · Attempt part e only.

(c) Estimate the number of units of alcohol that the patient has reduced their consumption by between March and August.

(1)

Estimation .....

(d) For the months of March and April, Rocio recorded the patient's alcohol consumption each week. Her results are shown in **Table 3**.

Week	Total alcohol consumption in units
March Week 1	79
March Week 2	77
March Week 3	72
March Week 4	70
April Week 1	79
April Week 2	63
April Week 3	45
April Week 4	34

Table 3

Calculate the mode for the units of alcohol consumed by the patient using **Table 3**.

(1)

Mode .....

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

P 7 5 5 2 4 A 0 1 0 4 8

6. Question 5 · part e · 2 marks

2024 JUN · Paper 02

Further practice · Attempt part e only.

DO NOT WRITE IN THIS AREA

(e) Calculate the median score for the units of alcohol consumed by the patient using **Table 3**.

(2)

Median .....

(f) On one day in March, the patient drank a total of 12.81924 units of alcohol.

Give the units of alcohol to **two** decimal places.

(1)

.....

(Total for Question 5 = 8 marks)

OLevels.COM SCHOOL

Working and answer